# Provisional Teacher Review

Acid–Base Concept Inventory — AI-Guided Virtual Science Lab (final-year research project)

## How to use this document

Please read each of the four items below alongside the suggested response. For each one, either agree with it, or write your own correction in the blank space provided — your own wording is what matters, not whether you agree with the draft. Then answer the one scope question and give your overall verdict near the end. Finally sign and date the last page.

A typed full name plus a typed or handwritten date in the signature block counts as your confirmation (a "virtual signature") — a photo or scan of the signed page, sent back to the researcher, is enough. Nothing here commits you to anything beyond confirming whether these four curriculum points and the overall inventory are reasonable for Grade 10/11 students.

## Four items for your review

### 1. Q4 / P4 — order of adding acid, indicator, and base in a titration

**P4.** A student wants to find the exact point at which an acid has been neutralised by a base, using phenolphthalein. In what order should they combine the three liquids?

(A) acid, then the indicator, then slowly add the base

(B) acid, then the base, then the indicator

(C) the base, then the indicator, then the acid

(D) it makes no difference — add all three at once

*Reason:*

(i) the indicator must already be in the acid so it can show the moment the base is in slight excess

(ii) the order in which chemicals are added never changes the result of a reaction

(iii) the indicator must be added last so it is not used up

(iv) the base must be added first because it is the more dangerous chemical

**Q4.** A student wants to find the exact point at which a base has been neutralised by an acid, using methyl orange. In what order should they combine the three liquids?

(A) base, then the indicator, then slowly add the acid

(B) base, then the acid, then the indicator

(C) acid, then the indicator, then the base

(D) it makes no difference — add all three at once

*Reason:*

(i) the indicator must already be in the base so it can show the moment the acid is in slight excess

(ii) the order chemicals are added in never changes the result

(iii) the indicator must be added last so it is not used up

(iv) the acid must be added first because it is more dangerous

*Scoring key on file: P4 correct answer is (A, i); Q4 correct answer is (A, i) — for P4, option A is "acid → indicator → base"; for Q4, option A is "base → indicator → acid." The two items are intentionally mirrored, not asking for one single universal order.*

**Suggested response:** There is no single universal order for every titration. The sensible order depends on which solution is in the flask (gets the indicator first) and which is the titrant (added last, slowly). P4 and Q4 are already set up as mirror cases for this reason, so no single item should claim one fixed order is the only correct titration setup.

**Recommendation:** Keep both items as a mirrored pair; just confirm the wording makes clear which solution is in the flask and which is added from the burette, so students aren't tempted to answer from a memorised single order.

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## 2. Q7 / P7 — "the pH at the end-point is always exactly 7"

**P7.** "At the exact end-point of an acid–base titration, the pH is always 7." This statement is:

(A) never true

(B) only true for a strong acid with a strong base

(C) always true

(D) only true when an indicator is used

*Reason:*

(i) the end-point pH depends on whether the acid and the base are strong or weak

(ii) neutral always means pH 7, whatever the acid and base are

(iii) indicators only work at pH 7

(iv) titrations are designed to stop exactly at pH 7

**Q7.** "At the exact end-point of an acid–base titration the solution is always neutral, pH 7." This is:

(A) never true

(B) only true for a strong acid with a strong base

(C) always true

(D) only true without an indicator

*Reason:*

(i) the end-point pH depends on the strengths of the acid and the base

(ii) neutral is always pH 7 regardless of the chemicals

(iii) indicators only change at pH 7

(iv) titrations always stop at pH 7 by design

**Suggested response:** The strong/weak acid and base distinction, and the fact that a salt can be acidic, basic, or neutral depending on the strengths of the parent acid and base, is taught in Grade 11 Science (not restricted to A/L Chemistry) — so the underlying idea is within scope conceptually. However, the item should say "equivalence point" (the chemically equivalent amounts) rather than "end-point" (the indicator's observed colour change) — even in a strong/strong titration, the observed indicator end-point is not necessarily exactly pH 7.

**Recommendation:** Keep the concept, but replace "end-point" with "equivalence point" throughout both items and the scoring key.

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### 3. Q12 / P12 — "add acid to water, never water to acid"

**P12.** When making up a dilute acid from a concentrated one, the safe instruction is 'add acid to water, never water to acid.' A student says the order does not matter as long as the same amounts are used. This student is:

(A) wrong — the order affects safety

(B) right — only the amounts matter

(C) right — order never matters in chemistry

(D) wrong — you should never dilute acid at all

*Reason:*

(i) adding water to concentrated acid can release heat so fast that the mixture spits or boils

(ii) the order chemicals are combined in has no effect on the outcome

(iii) water and acid do not mix if added in the wrong order

(iv) diluting acid is always unsafe

**Q12.** A technician says that when diluting concentrated sulfuric acid the order ('acid into water') does not matter as long as the final concentration is the same. This is:

(A) wrong — the order affects safety

(B) right — only concentration matters

(C) right — order never matters

(D) wrong — sulfuric acid must never be diluted

*Reason:*

(i) pouring water onto concentrated acid can release heat fast enough to make it spit or boil

(ii) the order substances are combined in has no effect on the outcome

(iii) acid and water will not mix if combined in the wrong order

(iv) diluting sulfuric acid is always unsafe

**Suggested response:** This is a separate laboratory-safety rule, not the same idea as titration order. Diluting acid is about rapid, localised heating when water hits concentrated acid; titration order is about when the indicator must be present and that the titrant is added gradually. These are different reasons and probably shouldn't be scored as evidence for the same misconception as Q4/P4.

**Recommendation:** Keep P12/Q12 as safety-only items. Do not count them as evidence for the titration-order misconception (CHEM_001); if a third item is still needed for that code, replace them with another titration-specific item.

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### 4. Q13 / P13 — weak/strong neutralisation and the final pH

**P13.** Dilute ethanoic acid (a weak acid) is neutralised by exactly the right amount of sodium hydroxide (a strong base). The final solution is:

(A) a little basic (pH above 7)

(B) exactly pH 7

(C) a little acidic (pH below 7)

(D) strongly acidic

*Reason:*

(i) the salt formed (sodium ethanoate) reacts slightly with water to make the solution a little basic

(ii) any exact neutralisation gives a solution of pH 7

(iii) a weak acid can never be fully neutralised

(iv) sodium hydroxide always leaves the solution strongly basic

**Q13.** Dilute ammonia solution (a weak base) is neutralised by exactly the right amount of hydrochloric acid (a strong acid). The final solution is:

(A) a little acidic (pH below 7)

(B) exactly pH 7

(C) a little basic (pH above 7)

(D) strongly basic

*Reason:*

(i) the salt formed (ammonium chloride) reacts slightly with water to make the solution a little acidic

(ii) any exact neutralisation gives pH 7

(iii) a weak base can never be fully neutralised

(iv) hydrochloric acid always leaves the solution strongly acidic

**Suggested response:** Grade 11 students are taught the conceptual result that a salt can be acidic, basic, or neutral depending on the strengths of the parent acid and base — they are not expected to use the detailed A/L treatment of salt hydrolysis, equilibrium constants, or a numerical pH calculation. The current reason wording ("the salt reacts slightly with water") introduces the hydrolysis mechanism by name, which may be more than Grade 11 students have seen.

**Recommendation:** Keep both items, but reword the reason to Grade 11-level language without naming the mechanism, e.g. "a salt formed from a weak acid and a strong base gives a basic solution" (P13) and the mirrored weak-base/strong-acid statement for Q13.

## The one scope question that matters most

**“Is the strong/weak acid-base distinction, and the idea that the end-point pH isn't always exactly 7, part of what Grade 10/11 students are taught — or would a typical student have no way to know that?”**

✔️ Yes, it's taught at Grade 10/11 — keep this item (CHEM_003) in the study.

☐ No, it isn't — drop CHEM_003 from the study.

☐ Unsure / depends — note why:

## Overall verdict

✔️ Approved as written.

☐ Approved, with the changes I've noted above.

☐ Needs more work — what:

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## Sign-off

By signing below, I confirm I have read this document, that the responses and corrections above (including any I've added) reflect my own professional judgement, and that they may be used, with attribution to me as named below, as the expert-review step of this student's final-year research project.

**Teacher name:** Chathura.S.Abeysinghe

**School / institution:** B.Sc. Engineering (Hons ) University Of Moratuwa

**Subject(s) taught:** Chemistry And Physics

**Signature (or typed full name as a virtual signature):** 

**Date:** 2026.09.30